

Predicting Bitcoin Returns with Machine Learning

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Abstract: We assess the predictability of daily Bitcoin returns using a broad set of market, macro, sentiment, and on-chain features from 2013-2024. We benchmark linear (OLS, LASSO), dimensionality-reduction (PCA), tree ensembles (Random Forest, XGBoost), and deep learning (feed-forward NN, LSTM) in a rolling one-day-ahead forecasting scheme. We find that LASSO and XGBoost significantly outperform OLS and most alternatives on squared-error loss. We construct long/short and long/cash trading strategies to assess the economic value of the signals. A simple two-asset allocation that trades only on equity-market days attains a Sharpe ratio of about 2.18 with LASSO forecasts versus 1.83 for buy-and-hold BTC. These gains translate into economically meaningful directional accuracy and fee-aware strategy performance. We link technical innovation in forecasting to portfolio design under realistic constraints. Results are robust to window choice, transaction costs, and alternative risk scaling, suggesting that while returns are hard to predict, certain models can extract economically meaningful signals that survive transaction costs, outperform naïve benchmarks and inform the future role of cryptocurrencies in diversified portfolios.

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1. Introduction

Bitcoin (BTC), the first and most valuable cryptocurrency, remains a volatile asset. This volatility has attracted considerable attention from researchers, institutional investors, and systematic traders. Yet, whether BTC returns are predictable continues to be a contested question. A growing body of literature has examined the issue using diverse empirical designs; from traditional linear regressions to advanced machine learning (ML) approaches across varying horizons, predictors, and methodological frameworks. Despite this effort, no consensus has emerged either on the most effective modeling techniques or on the direction and persistence of predictability.

This lack of clarity matters for both academic research and practice. For academics, it raises questions about the informational efficiency of cryptocurrency markets and the transferability of traditional asset-pricing tools to this new setting. For practitioners, the answer has direct implications for portfolio construction, risk management, and the development of systematic trading strategies in digital assets. The rapid growth of BTC as both an investable asset and a laboratory for financial innovation makes this debate especially timely.

In this paper, we revisit Bitcoin return predictability using modern ML methods and an expansive feature set that extends beyond standard price and volume indicators. Unlike prior studies that typically focus on narrow predictor classes or short-term classification tasks, we conduct a unified benchmarking exercise on daily BTC returns. Our predictors span market microstructure and liquidity measures, macroeconomic and risk factor proxies, investor attention and sentiment, and on-chain activity.

Methodologically, we compare a wide spectrum of models, including linear methods (OLS, LASSO), dimensionality reduction techniques (Principal Components based regression), ensemble

approaches (Random Forest, Gradient Boosting), and deep learning architectures (Neural Networks, Long Short-Term Memory). We adopt a rolling, fully out of sample design to assess forecasting accuracy and link predictive performance directly to trading outcomes under realistic frictions.

Our central finding is that models relying on regularization (LASSO) and ensemble learning XGBoost (XGB) consistently outperform both OLS and deep learning architectures. These approaches deliver robust predictive improvements that translate into economically meaningful Sharpe ratios, highlighting that parsimonious and structured ML models can outperform more complex alternatives in the cryptocurrency domain.

We build on three main strands of the literature. The first is the growing body of work on applying machine learning to asset pricing. Studies such as Feng et al. (2018), Weigand (2019), Gu et al. (2020), Giglio et al. (2022), and Bagnara (2024) demonstrate that ML models, more specifically tree-based methods and neural networks, enhance the forecasting of cross-sectional stock returns. Our study builds on this empirical ML setting but extends it to the time-series prediction of cryptocurrency returns.

The second strand is the literature on crypto return forecasting. Prior studies, including Chen (2023), Jaquart et al. (2021), Akyildirim et al. (2021), Liu et al. (2023), Wu et al. (2024), and Qureshi et al. (2025), apply ML methods to predict BTC prices across various horizons, from intraday to monthly, typically relying on relatively narrow sets of explanatory variables. We advance this line of research by incorporating a broader and more diverse set of features and by systematically comparing a wide spectrum of ML models in a unified, daily-frequency setting, which facilitates clearer benchmarking. Related work also evaluates Bitcoin in portfolios, documenting diversification gains and episodic hedge/safe-haven behavior relative to equities and

gold (Kajtazi & Moro, 2019; Guesmi et al., 2019; Klein et al., 2018; Mariana et al., 2021; Urquhart & Zhang, 2019). We connect these strands by showing that time-varying predictability can be operationalized via a risk-scaled allocation that rebalances only on equity-market days, improving portfolio outcomes under realistic frictions.

The third strand relates to the strategic evaluation of ML in finance. Studies such as Jiang and Liang (2017), Gort et al. (2022), and Cakici et al. (2024) examine the trading implications of ML models in the context of cross-sectional crypto returns. We contribute by focusing specifically on daily BTC returns and evaluating the performance of long/short and long/cash trading strategies, explicitly accounting for transaction costs and other realistic frictions.

Taken together, these contributions position our study as the first unified benchmarking of modern ML methods for daily Bitcoin return prediction, linking predictive accuracy to trading performance in a manner that is both methodologically rigorous and practically relevant. We link technical innovation in forecasting (regularized and tree-based ML) to implementable portfolio design for a 24/7 asset traded by investors who typically rebalance on equity-market days. First, we show that modest but robust out of sample gains from LASSO and XGBoost translate into fee-aware economic value under tight exposure bounds. Second, we detail the implementation choices that determine whether statistical gains are harvestable by aligning BTC to equity-day decision times (with weekend/holiday compounding), risk-scaling by recent BTC volatility, and strict box-constraints. Third, we evaluate a simple allocation that coexists with traditional assets and respects operational constraints post-ETF adoption. Together, the evidence speaks directly to the future role of cryptocurrencies in portfolios.

2. Literature Review

Early studies on Bitcoin sought to clarify its role, usefulness, and fundamental nature as a novel digital asset. Much of this work characterized Bitcoin as primarily speculative (Glaser et al., 2014; Geuder et al., 2019), often questioning whether it possessed any intrinsic value (Kristoufek, 2013; Cheah and Fry, 2015). A subsequent stream of research examined market efficiency, with evidence suggesting that Bitcoin has become progressively more efficient over time (Urquhart, 2016; Kinader and Papavassiliou, 2021). Nevertheless, disagreement about its classification persists. For instance, Dyhrberg (2016) views Bitcoin as a hybrid commodity that combines features of gold and fiat currencies, whereas Baur et al. (2018) argue that Bitcoin exhibits characteristics distinct from both gold and the U.S. dollar. Klein et al. (2018) further conclude that Bitcoin behaves unlike any other traditional asset class.

Debate has also centered on Bitcoin's function as a store of value and potential safe haven. Several studies report that Bitcoin can act as a short-term safe haven or diversification instrument under certain conditions (Demir et al., 2018; Wang et al., 2019; Kostika and Laopodis, 2019; Bouri, 2020; Mariana et al., 2021). Morillon (2022) argues that Bitcoin may serve as an effective hedge against expansionary monetary policies.

At the same time, significant skepticism remains. Bouoiyour and Selmi (2015) dismiss Bitcoin as "speculative foolery," while Schilling and Uhlig (2019) argue that it is a worthless medium of exchange vulnerable to manipulation. Other work highlights the growing influence of retail investors and social media on cryptocurrency markets (Urquhart, 2018; Aalborg et al., 2019), situating Bitcoin within broader phenomena of retail-driven market dynamics, as illustrated by the cases of GameStop and AMC in equity markets (Aloosh et al., 2022; Anand and Pathak, 2022; Chacon et al., 2023; Kogan et al., 2024).

These debates naturally led to inquiries into the valuation of digital assets such as Bitcoin. Some studies anchor valuation in Bitcoin's technical features and production costs (Dwyer, 2015; Hayes, 2019), while others incorporate measures of supply, demand, and market attractiveness (Ciaian et al., 2016). Alternative approaches include modeling scarcity via the stock-to-flow framework (Chacon and Morillon, 2023), capturing transactional demand (Cong and Li, 2021), and grounding valuation in adoption-based anchors (Bolt and Van Oordt, 2020).

While the early literature primarily debated Bitcoin's nature, value, and role in the financial system, a subsequent line of research has shifted attention toward the dynamics of Bitcoin returns. A central question is whether Bitcoin's price movements can be forecasted or whether they follow a pattern consistent with market efficiency. Early attempts relied on traditional econometric techniques, exploring drivers such as macroeconomic variables, risk factors, and investor sentiment. More recently, the rapid development of machine learning has opened new avenues for forecasting by allowing researchers to flexibly incorporate high-dimensional feature sets and capture complex, nonlinear relationships. This has given rise to a growing literature on return predictability in cryptocurrencies, with studies experimenting across horizons, model classes, and data inputs. Yet, despite this surge of interest, consensus remains elusive: findings vary widely depending on methodology, predictor choice, and evaluation design.

Returns predictability in the cryptocurrency space and the profitability of potential trading strategies based on such predictability remains an open and unsettled question. While numerous studies have attempted to address it, no consensus has emerged. As emphasized by Wu et al. (2024) in their review of the ML crypto literature, existing research varies widely in focus, methodology, and evaluation design, making it difficult to compare findings or establish reliable patterns. Studies

differ not only in their choice of models and horizons but also in what they attempt to forecast (price, returns, volatility, or trading signals) and in the variables and success metrics employed.

Early contributions investigated the predictability of Bitcoin and other cryptocurrency prices and returns using traditional econometric tools, often extended to altcoins (Sovbetov, 2018; Nguyen et al., 2019; Liu et al., 2021). Some examined derivatives markets, highlighting the informational role of futures and options (Corbet et al., 2018; Kim et al., 2020; Akyildirim et al., 2020; Nagula and Alexakis, 2022). Others focused on volatility prediction, applying GARCH-type models or hybrids with ML techniques such as LSTM networks (Garcia Medina and Aguayo-Moreno, 2024), with mixed evidence regarding accuracy gains.

A second group of studies has concentrated on directional return forecasting, using a wide range of economic, financial, risk, sentiment-based, and risk-based predictors. For example, Klein et al. (2018) and Jareno et al. (2020) explore gold prices as drivers, Demir et al. (2018) examine economic policy uncertainty, Aysan et al. (2019) incorporate geopolitical risks, and Corbet et al. (2020) study the influence of macroeconomic news. Other work emphasizes volatility and trading volume (Balcilar et al., 2017; Katsiampa, 2017) as potential predictors, whereas Liu and Tsyvinski (2021) use crypto-specific factors to predict returns, again with heterogeneous results.

More recent research has turned toward modern ML methods, though without converging on best practices. Panagiotidis et al. (2018) employ LASSO to identify relevant predictors, while Huang et al. (2019) use a range of technical indicators to forecast returns. Neural networks and deep learning architectures, including recurrent models such as LSTM, have also been applied at intraday to monthly horizons (Jaquart et al., 2021; Lazovic et al., 2023, Garcia Medina & Aguayo-Moreno, 2024; Bouteska et al., 2024), sometimes matching or surpassing simpler benchmarks. However, overall performance remains context-dependent, varying with time horizon, features,

and evaluation protocol as hybrid statistical/ML approaches can reshuffle rankings (Nagula & Alexakis, 2022; He et al., 2024). Situated at the intersection of digital assets and data assets, we combine market, macro, sentiment, and on-chain information with modern ML to extract implementable signals for Bitcoin.

Taken together, this literature highlights both the promise and the limitations of forecasting in cryptocurrency markets. Across studies, there is little agreement on which variables matter, on what horizon predictability is strongest, or even on the precise target of prediction (returns, prices, volatility, or volume). Moreover, methodological heterogeneity and differences in evaluation criteria make it difficult to compare results across papers. Consequently, the field lacks a systematic benchmarking effort that evaluates competing ML models in a consistent framework using a broad and economically motivated feature set.

To summarize, researchers have approached crypto predictability with diverse models across heterogeneous horizons and targets, complicating clear inference. At intraday/high frequencies, Bouri et al. (2022), Wen et al. (2022) and Pourrezaee and Hajizadeh (2024) show that models leveraging rich intraday features can yield profitable trading strategies or superior forecasts for non-return targets. With a daily frequency, studies using regularized LASSO methods (Ciganovic and D'amario, 2024), hybrid vs classical vs deep learning comparisons over multi-day ahead horizons (AlMadany et al., 2024), and studies of volatility forecasting with ML vs GARCH models (Huang et al., 2024) reveal that simpler/regularized models sometimes outperform more complex ones depending on predictor choice, sample period, and evaluation metric. At daily horizons, work spans regularized linear methods, tree/ensemble methods, and high-dimensional technical-indicator models; results depend heavily on predictor sets and evaluation design (Panagiotidis et al., 2018; Huang et al., 2019). At weekly or longer horizons, studies often

emphasize macro/uncertainty factors (EPU, geopolitical risk) and cross-asset links (gold), again yielding non-uniform conclusions across samples and regimes (Aysan et al., 2019; Cheng and Yen, 2020; Jareno et al., 2020; He et al., 2024). Target variables also differ: some forecast spot prices/returns (including directional accuracy), others focus on volatility with GARCH-family models and replications showing sensitivity to specification and sample windows; a parallel thread analyzes derivatives (futures/options) for price discovery and predictability (Katsiampa, 2017; Akyildirim, 2020). Collectively, these choices produce non-comparable success metrics and conflicting takeaways, reinforcing the need for a unified, daily-frequency benchmark with economically motivated features and out-of-sample trading evaluation, the gap this paper addresses.

This fragmentation underscores the need for a systematic and unified evaluation of Bitcoin return predictability. Our paper addresses this gap by benchmarking a broad set of machine learning models on daily BTC returns using an extensive range of economically motivated predictors, thereby providing a consistent framework to assess both forecasting performance and trading relevance.

3. Data and Features

Although Bitcoin price data is available from the earliest days of trading in January 2009, the inexistent market and absence of collected controls and predictive variables makes early data unexploitable. In an attempt to be exhaustive, we gather over 150 relevant variables, all with various sources and availabilities. Consequently, we use daily data from January 2013 through December 2024, the earliest period for which a broad set of market, macro, sentiment, and on-chain variables is reliably available. Market indicators include variables derived from trading and market data such as market cap, returns based measures, ranges, liquidity (e.g., Amihud style illiquidity),

volume and volatility-based measures, and price-rule indicators (e.g., moving-average crossovers). Macro proxies comprise the S&P 500 return, VIX, gold, the U.S. Dollar Index, Treasury yields, and balance-sheet aggregates. Sentiment indicators include variables capturing interest and attention in Bitcoin, including a BTC Google trend index and the Bitcoin Fear and Greed index as these have been shown to have some predictability (Ben Hamadou et al. 2025). On-Chain indicators include various variables capturing the health, activity and functionality of the Bitcoin blockchain such as exchange flows and reserves, address and transaction counts, miner activity, age-of-supply measures (e.g., CDD, MCDA, SCA), and valuation ratios (e.g., NVT, stock-to-flow).¹

Beyond variable construction, timing around Bitcoin’s issuance cycles is economically salient. *Figure 1* plots the BTC level and log price with vertical markers at the four halving events and shaded bull/bear regimes for exposition. The halvings are deterministic protocol changes that reduce the block subsidy roughly every four years and, hence, the flow supply of new BTC. Because this schedule is known in advance, incorporating days since halving (and a halving-day indicator) does not introduce look-ahead; it simply proxies for slow-moving supply pressure and the investor narratives that tend to build around these events. We treat *days since halving* as a continuous on-chain/protocol variable alongside our other predictors. Importantly, the green/red shading in Figure 1 is ex post and purely descriptive, we do not use any ex-post regime labels in model training. All variables are constructed at time t to forecast r_{t+1} ; scaling and any log or differencing transforms use only information in the training window to avoid look-ahead.

¹ All series are timestamped in UTC. Bitcoin prices use the 23:59 UTC close (the convention adopted by most data providers). Equity and macro variables follow their respective local market closes and are aligned to the next UTC day when constructing the daily dataset. All predictors are lagged one day so that only information available by the investor’s decision time is used.

$$E(R_{t+1}) = f(X_t) \quad (0)$$

Table 1 reports summary statistics for a representative subset in each category. For sake of space and brevity we do not define all of the variables in this section, but the list of variables used as well as its source and computation is defined in Appendix A.

4. Models and Methodology

We compare a suite of forecasting models across four families of supervised learning algorithms: linear models, dimensionality reduction techniques, tree-based ensemble methods, and deep learning architectures. Each model is chosen based on its prevalence in financial machine learning literature and its ability to handle varying data properties such as nonlinearity, high dimensionality, and temporal dependencies.

4.1 Linear Model

Linear models assume a linear relationship between the input variables X and the target variable y . We consider two variants, the basic Ordinary Least Squares (OLS) and Least Absolute Shrinkage and Selection Operator (LASSO). OLS provides a baseline model often used in the literature, but it can suffer from overfitting and multicollinearity in high dimensional settings. LASSO adds a penalty to enforce sparsity in the coefficients performing variable selection to reduce overfitting. It is particularly suited to our setting given the large number of predictor variable we test:

4.1.1 Ordinary Least Squares (OLS)

$$\hat{\beta}^{OLS} = \arg \min_{\beta} \frac{1}{n} \sum_t (y_t - x_t^T \beta)^2 \quad (1)$$

4.1.2 Least Absolute Shrinkage and Selection Operator (LASSO):

$$\hat{\beta}^{LASSO} = \arg \min_{\beta} \frac{1}{n} \sum_t (y_t - x_t^\top \beta)^2 + \lambda \|\beta\|_1 \quad (2)$$

4.2 Dimensionality Reduction

Another way to address dimensionality and multicollinearity is to apply Principal Component Analysis (PCA). PCA projects variable vector X onto a lower-dimensional space via orthogonal transformation:

$$Z = \tilde{X}W_k \in \mathbb{R}^{n \times k}. \quad (3)$$

where W contains the top k eigenvectors of the covariance matrix X . We then apply OLS regression on the transformed features Z .

4.3 Tree-Based Models

Tree-based models offer flexibility in modeling nonlinear interactions and are especially robust to noisy data, a trait particularly useful in our setting. Random Forest (RF) averages B decision trees $\{T_b\}$ trained on bootstrapped samples. Extreme Gradient Boosting (XGB) builds tree sequentially to correct errors from previous trees.

4.3.1 Random Forest (RF)

$$\hat{y}^{RF}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x) \quad (4)$$

Each tree T_b partitions the space into leaves $\{R_{bl}\}_{l=1}^{L_b}$ by minimizing within-node squared error. Each tree T_b is a CART grown on a bootstrap sample; at each split we consider a random subset of features (m_{try} ; default $\sqrt{p}$). Predictions are the in-leaf mean, and the forest prediction is the average across trees.

4.3.2 Extreme Gradient Boosting (XGB):

For every new iteration t , a new tree $f_t(x)$ minimizes:

$$L^{(t)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (5)$$

We use squared-error loss $\ell(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$ and $\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^T w_j^2$, where T is the number of leaves and w_j the leaf weights.

4.4 Deep Learning Models

Deep learning models learn hierarchical representation of features. We use Feed Forward Neural Network (NN) which maps input to output via multiple hidden layers, and Long Short-Term Memory (LSTM) that are designed for sequential data, with memory cells updating through input, forget and output gates.

4.4.1 Feed Forward Neural Network (NN):

$$\hat{y} = f(x; \theta) (x) = \phi_L(W_L \phi_{L-1}(\dots \phi_1(W_1 x + b_1) \dots) + b_L) \quad (6)$$

Hidden layers use ReLU activations (ϕ_L) and a linear output for regression; parameters $\theta = \{W_L, b_L\}$ are fitted by minimizing MSE with optional L_2 weight decay.

4.4.2 Long Short-Term Memory (LSTM):

$$\begin{aligned} i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), & f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o) \\ \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c), & c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \\ h_t &= o_t \odot \tanh(c_t) \end{aligned} \quad (7)$$

Here, $\sigma(\cdot)$ denotes the logistic function and $\odot$ denotes element-wise multiplication, as used in the gating equations of Equation (7). For each day t , we feed a fixed window of recent observations

into the LSTM and apply the recursions across that window to obtain the final hidden state h_t . The model then produces the one-day-ahead return forecast directly from this final state through a linear output layer.

4.5 Hyperparameters

At each rolling step, we tune hyperparameters inside the current training window using time-ordered cross-validation. The window is split into contiguous folds where each validation fold occurs after its training fold. All preprocessing (standardization and PCA when used) is fit on the training fold only and applied to the validation fold to prevent leakage. We select settings by minimizing validation MSE (ties favor sparser settings), then refit the chosen model on the full training window and issue a one day ahead forecast.² LASSO uses 5 blocked folds over a logarithmic grid of α ; Random Forest and XGBoost use *RandomizedSearchCV* with a time-ordered 2-fold split; NN and LSTM use KerasTuner RandomSearch with an in window chronological 80/20 validation split and early stopping. PCA retains the minimum number of components that explains 90% of variance, recomputed within each window. All splits preserve chronology (no shuffling), and random seeds are fixed for reproducibility. Full search spaces and hyperparameters settings are summarized in Appendix A.

4.6 Evaluation Methodology

We use a walk-forward expanding framework. At each day t , the model is estimated recursively on all data available up to $t-1$ (the window grows by one observation each step), a one-day-ahead

² For day t , hyperparameters are chosen only from data up to $t-1$ via the inner CV splits. After selection, we refit on the entire window up to $t-1$ including observations that temporarily served as validation in CV, and then produce the forecast for t . Thus, we never use information from t or later when fitting, while still exploiting all information available as of $t-1$.

forecast for t is produced, and then day t is appended to the estimation sample for the next iteration.³ This mirrors realistic trading conditions in which the investor updates decisions using only information available at the time. Our base training sample consists in the ten-year period ranging from 1/1/2013 to 12/31/2022. Our base Test windows ranges from 1/1/2023 to 12/31/2024 (a) and 1/1/2022 to 12/31/2024 (b). These two windows allow to investigate the robustness of results using different time periods (starting in January 2022 includes a good chunk of the bear market) all while keeping the training and test sample within reasonable boundaries. Model performance is then evaluated using several metrics like Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Mean Squared Relative Error (MSRE) to assess forecast precision. All evaluations are conducted out-of-sample (OOS) using rolling model retraining.

4.7 Portfolio evaluation

Bitcoin trades continuously, including weekends and holidays, whereas the equity market rebalances only on trading days. To make a fair, implementable comparison on the equity market's daily schedule, we aggregate Bitcoin's calendar-day returns between two consecutive equity closes into a single return used for that trading day. Let r_s^{BTC} denote the simple Bitcoin return on calendar day s , and let w_t be the set of calendar days since the previous equity close up to (and including) equity trading day t . The aligned Bitcoin return for day t is

$$r_t^{BTC \text{ aligned}} = \prod_{s \in w_t} (1 + r_s^{BTC}) - 1 \quad (8)$$

³ We prefer an expanding window for stability; a fixed-length rolling window is a natural alternative and yields qualitatively similar results.

This construction means we do not trade on non-equity days: positions are held through weekends and holidays, and all intervening Bitcoin performance is geometrically compounded into the next equity trading day. The equity return for day t is the standard one-day equity return for that trading day.

On each trading day, we construct a two asset portfolio that consists of Bitcoin and the CRSP value-weighted index (i.e., equities). First, we construct unscaled weights for each asset as follows:

$$w_t^{BTC,unscaled} = \frac{\widehat{r_{BTC_{t+1}}}}{\sigma_t^{2,BTC}}, \quad w_t^{Mkt,unscaled} = \frac{\bar{r}_{Mkt,t+1}}{\sigma_t^{2,Mkt}} \quad (9)$$

Here, $\widehat{r_{BTC_{t+1}}}$ is the predicted return from the machine learning models, $\sigma_t^{2,BTC}$ and $\sigma_t^{2,Mkt}$ are variance of returns on Bitcoin and the CRSP value-weighted market index using lagged 60-day rolling window of returns, and $\bar{r}_{Mkt,t+1}$ represents the mean returns on the CRSP value-weighted index using the last 60 days.

Since the strategy only holds these two assets, we normalize the unscaled weights such that their sum equals one⁴:

$$w_t^{BTC} = \frac{w_t^{BTC,unscaled}}{(w_t^{BTC,unscaled} + w_t^{Mkt,unscaled})}, \quad w_t^{Mkt} = 1 - w_t^{BTC} \quad (10)$$

We further apply realistic long-short bounds and constrain the weights between -50% and 150% (Rapach et al., 2016; Kothari and O'Doherty, 2023). The daily strategy returns are then computed as the weighted sum of returns on Bitcoin and the CRSP value-weighted index.

⁴ If the unscaled sum equals zero, the denominator is replaced with one to avoid division by zero while yielding neutral weights.

5. Empirical Results

Table 2 compares models on conventional loss metrics. Over 2023-24, XGB and LASSO achieve the lowest RMSE/MAE, while unpenalized OLS and PCA based baselines underperform. The relative-error metrics (MSRE, MAPE) broadly corroborate the extremes and also identify XGB as best and OLS as worst. They disagree, however, on intermediate ranks because their denominators magnify days when the actual return is near zero (this explains some very large MSRE (e.g. LSTM $\sim 3,300$) as a few near-zero-return days cause the error to blow up even when absolute errors are modest).⁵ Accordingly, we base model ordering on RMSE/MAE and treat MSRE/MAPE as secondary diagnostics. The strong out-of-sample showing of gradient-boosted trees is reminiscent of Bouteska et al. (2024), who also report competitive performance for GBM models in daily crypto settings. These results are robust to alternative rolling windows and to excluding the noisiest predictors.

In *Table 3*, we report out of sample R^2 (R_{OOS}^2) which provide a model-free measure of incremental forecasting power relative to the historical mean. Although magnitudes are modest (which is typical of daily-return predictability in crypto models), Gradient Boosting and LASSO are the leading models and deliver positive R_{OOS}^2 throughout the evaluation window. Tree-based methods capture mild nonlinearities, while LASSO’s shrinkage stabilizes the linear fit in a high-dimensional setting. Interestingly, Deep learning architectures (Feed-Forward NNs and LSTMs) do not dominate in this daily horizon suggesting that most of the exploitable structure is cross-sectional across features rather than deeply temporal.

⁵ MSE/MAE are primary metrics. $MSRE = \frac{1}{T} \sum (e_t/y_t)^2$ and $MAPE = \frac{1}{T} \sum |e_t/y_t|$ can over-weight days with $|y_t| \approx 0$, which may impact rankings. We therefore interpret MSRE/MAPE cautiously.

Table 4 reports sign-prediction performance relative to the 50.62% share of up days in the sample. XGB and LASSO deliver the strongest directional accuracy with 56.36% and 56.22% correct, with p-values both significant at the 1% level. Their hit rates are balanced across states (XGB: 55.70% on up days, 57.39% on down days; LASSO: 55.41% and 57.62%), despite a long tilt in predicted signs (61.15% and 63.20% positives vs. 50.62% in the data). PCA and LSTM reach 53.21% correct with $p = 0.044$ indicating modest but statistically significant directional performance at the 5% level. OLS (51.44%, $p=0.2297$) is indistinguishable from chance, while NN (49.38%, $p=0.6442$) and RF (48.70%, $p=0.7703$) underperform. The weaker models are characterized by substantial imbalance (OLS and NN predict positives 69.08% and 74.69% of the time), which depresses the “correct negative” rate and overall accuracy. In contrast, the top models maintain comparatively balanced performance across up and down days further confirming our initial diagnostic.

Before turning to the allocation exercise, we benchmark the most transparent way to “invest with” the forecasts: a sign-based trading rule. *Figure 2* plots cumulative returns from two implementations. The long/short rule (1/−1) goes long BTC when the model predicts a positive next-day return and short when it predicts a negative one; the long/cash rule (1/0) goes to cash on negative signals. We report both gross (no fees) and net) alongside buy-and-hold BTC as a reference (yellow dashed line). We apply a proportional transaction cost of 10 bps per trade (per leg) to spot BTC, in line with major exchange schedules (e.g., Binance standard 0.1%) and prior empirical findings (Yang, 2025). Signals are produced in a rolling, walk-forward manner and positions are set using only information available at $t-1$.

Two patterns stand out. First, without fees the better statistical models (LASSO and XGB) translate their higher sign accuracy into faster compounding, with PCA competitive

and OLS/RF/LSTM generally lagging. Second, after applying a 10 bps per-leg fee, the long/short rule still outperforms long/cash for the stronger models. This indicates that the average return on days flagged as negative is sufficiently below zero to offset the extra trading from sign reversals. Long/cash remains the more cost-resilient implementation and would dominate under higher costs, weaker downside accuracy, or very frequent signal flips, but in our sample the downside hit rate of LASSO and XGB is strong enough that shorting adds value. The ranking across models mirrors the error-metric and DM results: regularization and ensembling deliver steadier, more actionable signals, and the net performance premium is most visible in higher-volatility episodes where correct downside calls matter most. Overall, the figure shows that even modest predictive skill can be converted into economically meaningful outperformance with simple, transparent rules.

In *Table 5* we formally compare models using Diebold–Mariano tests (squared-error loss; positive statistics favor the column model). Gradient boosting (XGB) and LASSO dominate the field. XGB significantly outperforms OLS (4.24), RF (3.82), and NN (4.11), and also beats LSTM at the 10% level (1.86). Differences versus LASSO (0.18) and PCA (0.82) are not statistically significant. Likewise, LASSO improves upon OLS (4.22), RF (3.36), NN (3.22), and edges LSTM at the 10% level (1.90), while remaining statistically on par with PCA (0.82) and XGB (-0.18). Notably, PCA is not a mere baseline: it beats OLS (4.11), RF (2.79), and NN (2.76). The most striking pattern is that OLS is uniformly inferior to every ML alternative. Every entry in the OLS row is large and positive ($DM > 3.2$), indicating significantly higher forecast loss against each competing method. RF, NN, and LSTM are generally weaker than the top two and often indistinguishable from one another. Overall, the DM evidence reinforces the error-metric ordering: regularization (LASSO) and ensembling (XGB) extract stable predictive structure that simple linear regression and most black-box alternatives fail to capture.

Next, we look at possible trading strategies and implementations. In Table 6, we evaluate a two-asset allocation that trades only on equity-market days. Because BTC trades 24/7, we aggregate BTC’s non-trading-day returns between two equity closes into a single BTC return for that trading day (geometric compounding of weekend/holiday returns and the current day’s BTC return, see equation 8). Each equity day we set the BTC sleeve’s raw weight proportional to the model’s forecasted next-day BTC return scaled by recent risk, apply box constraints $[-0.5, 1.5]$, and renormalize the two sleeves to sum to 1 (see equation 9). Sharpe ratios are computed on daily strategy returns (net of the risk-free) and annualized.⁶

The strategy based on LASSO forecasts achieves the highest risk-adjusted performance with a Sharpe ratio of 2.18. It outperforms both buy-and-hold BTC (Sharpe = 1.83), and allocations using alternative ML forecasts including XGB (1.72). The superiority of LASSO is consistent with our earlier statistical results: LASSO’s regularization stabilizes signals, improving the signal-to-risk ratio underlying the weight rule and translating small R^2_{OOS} gains into economically meaningful improvements. Results are robust to EWMA vol estimates, alternative box constraints, and fee deductions; using each asset’s own variance (or a full covariance step) produces similar rankings and avoids inadvertent scaling by BTC volatility in the equity sleeve. For 2023–24, the LASSO-based allocation achieves the highest Sharpe (2.18), outperforming buy-and-hold BTC (1.83) and the XGB allocation (1.72). This echoes the statistical results: LASSO’s shrinkage stabilizes signals, improving the signal-to-risk ratio that drives the weight rule, so even modest R^2_{OOS} gains convert into economically meaningful performance.

⁶ We annualize daily Sharpe by $\sqrt{252}$ to match equity trading days.

In the longer 2022–24 window, Sharpe ratios compress with the inclusion of the 2022 bear market; XGB (0.45) edges buy-and-hold BTC (0.45), while LASSO remains positive (0.33). The ordering is consistent with trees capturing some nonlinearity during drawdowns, whereas LASSO’s advantage is strongest in the 2023–24 regime. Across robustness checks (EWMA vol estimates, alternative box constraints, and using each sleeve’s own variance or a full covariance step), rankings are qualitatively unchanged, and scaling by the equity sleeve’s own risk avoids inadvertently keying that sleeve off BTC’s volatility as in our baseline specification.

There are several implications for portfolio design. First, the allocation is deliberately simple: risk-scaled BTC weight with tight bounds, rebalanced only on equity-market days. Second, implementability is high post-ETF:⁷ daily equity-close rebalancing harvests signal without requiring 24/7 execution. In practice, a conservative policy that limits BTC risk to 10–20% of total on signal-active days and otherwise defers to equities captures most of the Sharpe gains we document, with low operational burden.

6. Conclusion

We reassess daily Bitcoin return predictability using a broad set of market, macro, sentiment, and on-chain predictors, benchmarking linear, tree-based, and deep models in a walk-forward expanding design. Regularization and esembling matter: LASSO and SGBoost consistently outperform OLS and most alternatives on squared error loss and directional accuracy.

Crucially, these statistical gains convert into economic value when mapped to a simple, implementable allocation that rebalances only on equity-market days and respects realistic constraints and fees. In 2023–24, the LASSO-based allocation achieves a materially

⁷ The first Futures-based ETF (BITO) was launched on October 19, 2021, and the first Spot based ETF was launched on January 10, 2024 (11 different ETFs).

higher Sharpe than buy-and-hold BTC, with competitive performance in samples that include the 2022 bear market. Results are robust to volatility estimators, box-constraint ranges, and transaction-cost assumptions. Evidence on time-varying co-movement with equities clarifies why a risk-scaled, bounded allocation can improve mixed-asset portfolios despite volatile unconditional returns. Overall, signals are modest but usable; implementation details determine whether they are harvestable. Extensions include multi-asset crypto sets, options-implied information, and regime-aware or meta-model combinations.

References

- Akyildirim, Erdinc, Ahmet Goncu, and Ahmet Sensoy. "Prediction of cryptocurrency returns using machine learning." *Annals of Operations Research* 297, no. 1 (2021): 3-36.
- Akyildirim, Erdinc, Shaen Corbet, Paraskevi Katsiampa, Neil Kellard, and Ahmet Sensoy. "The development of bitcoin futures: Exploring the interactions between cryptocurrency derivatives." *Finance Research Letters* 34 (2020): 101234.
- AlMadany, Nehal N., Omar Hujran, Ghazi Al Naymat, and Aktham Maghyereh. "Forecasting cryptocurrency returns using classical statistical and deep learning techniques." *International Journal of Information Management Data Insights* 4, no. 2 (2024): 100251.
- Aloosh, Arash, Samuel Ouzan, and Syed Jawad Hussain Shahzad. "Bubbles across meme stocks and cryptocurrencies." *Finance Research Letters* 49 (2022): 103155.
- Anand, Abhinav, and Jalaj Pathak. "The role of Reddit in the GameStop short squeeze." *Economics Letters* 211 (2022): 110249.
- Amihud, Yakov. "Illiquidity and stock returns: cross-section and time-series effects." *Journal of financial markets* 5, no. 1 (2002): 31-56.
- Aysan, Ahmet Faruk, Ender Demir, Giray Gozgor, and Chi Keung Marco Lau. "Effects of the geopolitical risks on Bitcoin returns and volatility." *Research in International Business and Finance* 47 (2019): 511-518.
- Balcilar, Mehmet, Elie Bouri, Rangan Gupta, and David Roubaud. "Can volume predict Bitcoin returns and volatility? A quantiles-based approach." *Economic Modelling* 64 (2017): 74-81.
- Bagnara, Matteo. "Asset pricing and machine learning: A critical review." *Journal of Economic Surveys* 38, no. 1 (2024): 27-56.
- Baur, Dirk G., Thomas Dimpfl, and Konstantin Kuck. "Bitcoin, gold and the US dollar—A replication and extension." *Finance research letters* 25 (2018): 103-110.
- Ben Hamadou, Fatma, Taicir Mezghani, and Mouna Boujelbène Abbes. "Quantile-time-frequency risk spillover between investor attention, clean, and dirty cryptocurrency returns: FB Hamadou et al." *Risk Management* 27, no. 3 (2025): 10.
- Bolt, Wilko, and Maarten RC Van Oordt. "On the value of virtual currencies." *Journal of Money, Credit and Banking* 52, no. 4 (2020): 835-862.
- Bouoiyour, Jamal, and Refk Selmi. "What does Bitcoin look like?." *Annals of Economics & Finance* 16, no. 2 (2015).
- Bouri, Elie, Chi Keung Marco Lau, Tareq Saeed, Shixuan Wang, and Yuqian Zhao. "On the intraday return curves of Bitcoin: Predictability and trading opportunities." *International Review of Financial Analysis* 76 (2021): 101784.

Bouteska, Ahmed, Mohammad Zoynul Abedin, Petr Hajek, and Kunpeng Yuan. "Cryptocurrency price forecasting—a comparative analysis of ensemble learning and deep learning methods." *International Review of Financial Analysis* 92 (2024): 103055.

Cakici, Nusret, Syed Jawad Hussain Shahzad, Barbara Będowska-Sójka, and Adam Zaremba. "Machine learning and the cross-section of cryptocurrency returns." *International Review of Financial Analysis* 94 (2024): 103244.

Chacon, Ryan G., and Thibaut G. Morillon. "Dissecting the stock to flow model for Bitcoin." *Studies in Economics and Finance* 39, no. 3 (2022): 506-523.

Chacon, Ryan G., Thibaut G. Morillon, and Ruixiang Wang. "Will the reddit rebellion take you to the moon? Evidence from WallStreetBets." *Financial Markets and Portfolio Management* 37, no. 1 (2023): 1-25.

Chaim, Pedro, and Márcio P. Laurini. "Volatility and return jumps in bitcoin." *Economics Letters* 173 (2018): 158-163.

Cheah, Eng-Tuck, and John Fry. "Speculative bubbles in Bitcoin markets? An empirical investigation into the fundamental value of Bitcoin." *Economics letters* 130 (2015): 32-36.

Chen, Junwei. "Analysis of bitcoin price prediction using machine learning." *Journal of risk and financial management* 16, no. 1 (2023): 51.

Cheng, Hui-Pei, and Kuang-Chieh Yen. "The relationship between the economic policy uncertainty and the cryptocurrency market." *Finance Research Letters* 35 (2020): 101308.

Ciaian, Pavel, Miroslava Rajcaniova, and d'Artis Kancs. "The economics of BitCoin price formation." *Applied economics* 48, no. 19 (2016): 1799-1815.

Ciganovic, Milos, and Federico D'Amario. "Forecasting cryptocurrencies log-returns: a LASSO-VAR and sentiment approach." *Applied Economics* 56, no. 58 (2024): 8112-8138.

Cong, Lin William, Ye Li, and Neng Wang. "Tokenomics: Dynamic adoption and valuation." *The Review of Financial Studies* 34, no. 3 (2021): 1105-1155.

Cong, Lin William, George Andrew Karolyi, Ke Tang, and Weiyi Zhao. "Value premium, network adoption, and factor pricing of crypto assets." *Network Adoption, and Factor Pricing of Crypto Assets* (April 1, 2022) (2022).

Corbet, Shaen, Brian Lucey, Maurice Peat, and Samuel Vigne. "Bitcoin Futures - What use are they?." *Economics Letters* 172 (2018): 23-27.

Corbet, Shaen, Charles Larkin, Brian M. Lucey, Andrew Meegan, and Larisa Yarovaya. "The impact of macroeconomic news on Bitcoin returns." *The European Journal of Finance* 26, no. 14 (2020): 1396-1416.

de la Horra, Luis P., Gabriel de la Fuente, and Javier Perote. "The drivers of Bitcoin demand: A short and long-run analysis." *International Review of Financial Analysis* 62 (2019): 21-34.

Demir, Ender, Giray Gozgor, Chi Keung Marco Lau, and Samuel A. Vigne. "Does economic policy uncertainty predict the Bitcoin returns? An empirical investigation." *Finance Research Letters* 26 (2018): 145-149.

Dyhrberg, Anne Haubo. "Hedging capabilities of bitcoin. Is it the virtual gold?." *Finance research letters* 16 (2016): 139-144.

Feng, Guanhao, Jingyu He, and Nicholas G. Polson. "Deep learning for predicting asset returns." *arXiv preprint arXiv:1804.09314* (2018).

García-Medina, Andrés, and Ester Aguayo-Moreno. "LSTM–GARCH hybrid model for the prediction of volatility in cryptocurrency portfolios." *Computational Economics* 63, no. 4 (2024): 1511-1542.

Giglio, Stefano, Bryan Kelly, and Dacheng Xiu. "Factor models, machine learning, and asset pricing." *Annual Review of Financial Economics* 14, no. 1 (2022): 337-368.

Glaser, Florian, Kai Zimmermann, Martin Haferkorn, Moritz Christian Weber, and Michael Siering. "Bitcoin-asset or currency? revealing users' hidden intentions." *Revealing Users' Hidden Intentions* (April 15, 2014). ECIS (2014).

Gort, Berend Jelmer Dirk, Xiao-Yang Liu, Xinghang Sun, Jiechao Gao, Shuaiyu Chen, and Christina Dan Wang. "Deep reinforcement learning for cryptocurrency trading: Practical approach to address backtest overfitting." *arXiv preprint arXiv:2209.05559* (2022).

Geuder, Jan, Benjamin Kinateder, and Franz-Josef Kiss. "The role of Bitcoin in well-diversified portfolios." *Finance Research Letters* 28 (2019): 308–319.

Gu, Shihao, Bryan Kelly, and Dacheng Xiu. "Empirical asset pricing via machine learning." *The Review of Financial Studies* 33, no. 5 (2020): 2223-2273.

Guesmi, Khaled, Samir Saadi, Ilyes Abid, and Zied Ftiti. "Portfolio diversification with virtual currency: Evidence from bitcoin." *International Review of Financial Analysis* 63 (2019): 431-437.

He, Xiangyi, Yiwei Li, and Houjian Li. "Revolutionizing Bitcoin price forecasts: A comparative study of advanced hybrid deep learning architectures." *Finance Research Letters* 69 (2024): 106136.

Hou, Kewei, Haitao Mo, Chen Xue, and Lu Zhang. "An augmented q-factor model with expected growth." *Review of Finance* 25, no. 1 (2021): 1-41.

Huang, Jing-Zhi, William Huang, and Jun Ni. "Predicting bitcoin returns using high-dimensional technical indicators." *The Journal of Finance and Data Science* 5, no. 3 (2019): 140-155.

Huang, Zih-Chun, Ivan Sangiorgi, and Andrew Urquhart. "Forecasting Bitcoin volatility using machine learning techniques." *Journal of International Financial Markets, Institutions and Money* 97 (2024): 102064.

Jareño, Francisco, María de la O González, Marta Tolentino, and Karen Sierra. "Bitcoin and gold price returns: A quantile regression and NARDL analysis." *Resources Policy* 67 (2020): 101666.

- Jaquart, Patrick, David Dann, and Christof Weinhardt. "Short-term bitcoin market prediction via machine learning." *The journal of finance and data science* 7 (2021): 45-66.
- Jiang, Zhengyao, and Jinjun Liang. "Cryptocurrency portfolio management with deep reinforcement learning." In *2017 Intelligent systems conference (IntelliSys)*, pp. 905-913. IEEE, 2017.
- Kajtazi, Anton, and Andrea Moro. "The role of bitcoin in well diversified portfolios: A comparative global study." *International Review of Financial Analysis* 61 (2019): 143-157.
- Katsiampa, Paraskevi. "Volatility estimation for Bitcoin: A comparison of GARCH models." *Economics letters* 158 (2017): 3-6.
- Kim, Wonse, Junseok Lee, and Kyungwon Kang. "The effects of the introduction of Bitcoin futures on the volatility of Bitcoin returns." *Finance Research Letters* 33 (2020): 101204.
- Kinateder, Harald, and Vassilios G. Papavassiliou. "Calendar effects in bitcoin returns and volatility." *Finance Research Letters* 38 (2021): 101420.
- Klein, Tony, Hien Pham Thu, and Thomas Walther. "Bitcoin is not the New Gold—A comparison of volatility, correlation, and portfolio performance." *International Review of Financial Analysis* 59 (2018): 105-116.
- Kothari, Pratik, and Michael S. O'Doherty. "Job postings and aggregate stock returns." *Journal of Financial Markets* 64 (2023): 100804.
- Kristoufek, Ladislav. "BitCoin meets Google Trends and Wikipedia: Quantifying the relationship between phenomena of the Internet era." *Scientific reports* 3, no. 1 (2013): 3415.
- Kogan, Shimon, Igor Makarov, Marina Niessner, and Antoinette Schoar. "Are cryptos different? evidence from retail trading." *Journal of Financial Economics* 159 (2024): 103897.
- Lazović, Ivan, Bojan Đorđević, and Marija Lukić. "Bitcoin monthly return forecast: A comparison of ARIMA and multi layer Perceptron Artificial Neural Network." *International Review* 1-2 (2023): 149-162.
- Liu, Yukun, and Aleh Tsyvinski. "Risks and returns of cryptocurrency." *The Review of Financial Studies* 34, no. 6 (2021): 2689-2727.
- Liu, Yujun, Zhongfei Li, Ramzi Nekhili, and Jahangir Sultan. "Forecasting cryptocurrency returns with machine learning." *Research in International Business and Finance* 64 (2023): 101905.
- Llorente, Guillermo, Roni Michaely, Gideon Saar, and Jiang Wang. "Dynamic volume-return relation of individual stocks." *The Review of financial studies* 15, no. 4 (2002): 1005-1047.
- Mariana, Christy Dwita, Irwan Adi Ekaputra, and Zaäfri Ananto Husodo. "Are Bitcoin and Ethereum safe-havens for stocks during the COVID-19 pandemic?." *Finance research letters* 38 (2021): 101798.
- Morillon, Thibaut G. "Bitcoin's value proposition: shorting expansionary monetary policies." *Studies in Economics and Finance* 39, no. 1 (2022): 20-44.

Nagula, Pavan Kumar, and Christos Alexakis. "A new hybrid machine learning model for predicting the bitcoin (BTC-USD) price." *Journal of Behavioral and Experimental Finance* 36 (2022): 100741.

Nguyen, Thai Vu Hong, Binh Thanh Nguyen, Thanh Cong Nguyen, and Quang Quoc Nguyen. "Bitcoin return: Impacts from the introduction of new altcoins." *Research in International Business and Finance* 48 (2019): 420-425.

Panagiotidis, Theodore, Thanasis Stengos, and Orestis Vravosinos. "On the determinants of bitcoin returns: A LASSO approach." *Finance Research Letters* 27 (2018): 235-240.

Pourrezaee, Arash, and Ehsan Hajizadeh. "Forecasting bitcoin volatility and value-at-risk using stacking machine learning models with intraday data." *Computational Economics* (2024): 1-31.

Qureshi, Shavez Mushtaq, Atif Saeed, Farooq Ahmad, Asad Rehman Khattak, Sultan H. Almotiri, Mohammed A. Al Ghamdi, and Muhammad Shah Rukh. "Evaluating machine learning models for predictive accuracy in cryptocurrency price forecasting." *PeerJ Computer Science* 11 (2025): e2626.

Rapach, David E., Matthew C. Ringgenberg, and Guofu Zhou. "Short interest and aggregate stock returns." *Journal of Financial Economics* 121, no. 1 (2016): 46-65.

Sovbetov, Yhlas. "Factors influencing cryptocurrency prices: Evidence from bitcoin, ethereum, dash, litecoin, and monero." (2018): 1-27.

Urquhart, Andrew. "The inefficiency of Bitcoin." *Economics Letters* 148 (2016): 80-82.

Urquhart, Andrew. "What causes the attention of Bitcoin?." *Economics letters* 166 (2018): 40-44.

Weigand, Alois. "Machine learning in empirical asset pricing." *Financial Markets and Portfolio Management* 33, no. 1 (2019): 93-104.

Urquhart, Andrew, and Hanxiong Zhang. "Is Bitcoin a hedge or safe haven for currencies? An intraday analysis." *International Review of Financial Analysis* 63 (2019): 49-57.

Wen, Zhuzhu, Elie Bouri, Yahua Xu, and Yang Zhao. "Intraday return predictability in the cryptocurrency markets: Momentum, reversal, or both." *The North American Journal of Economics and Finance* 62 (2022): 101733.

Wu, Jingyang, Xinyi Zhang, Fangyixuan Huang, Haochen Zhou, and Rohtiash Chandra. "Review of deep learning models for crypto price prediction: implementation and evaluation." *arXiv preprint arXiv:2405.11431* (2024).

Yang, Dennis, and Qiang Zhang. "Drift-independent volatility estimation based on high, low, open, and close prices." *The Journal of Business* 73, no. 3 (2000): 477-492.

Yang, Ao. "Cryptocurrency Market Risk-Managed Momentum Strategies." *Finance Research Letters* (2025): 107879.

Figure 1: Bitcoin price, log price and halving dates

This figure shows the daily BTC price (top) and log price (bottom). Vertical black lines indicate the four protocol halving dates. Green (red) shading depicts ex-post bull (bear) regimes.



Figure 2: Cumulative returns from sign-based trading rules using model forecasts

Panels show cumulative returns for strategies driven by one-day-ahead model forecasts. Panel A: long/short (1/-1) with and without fees. Panel B: long/cash (1/0) with and without fees. The dashed line is buy-and-hold BTC. On each trading day, the position is set using the forecast sign from information available at $t-1$; fees are the same transaction-cost assumption used throughout the paper. Higher lines indicate better cumulative performance.

Panel A: 2023 - 2024

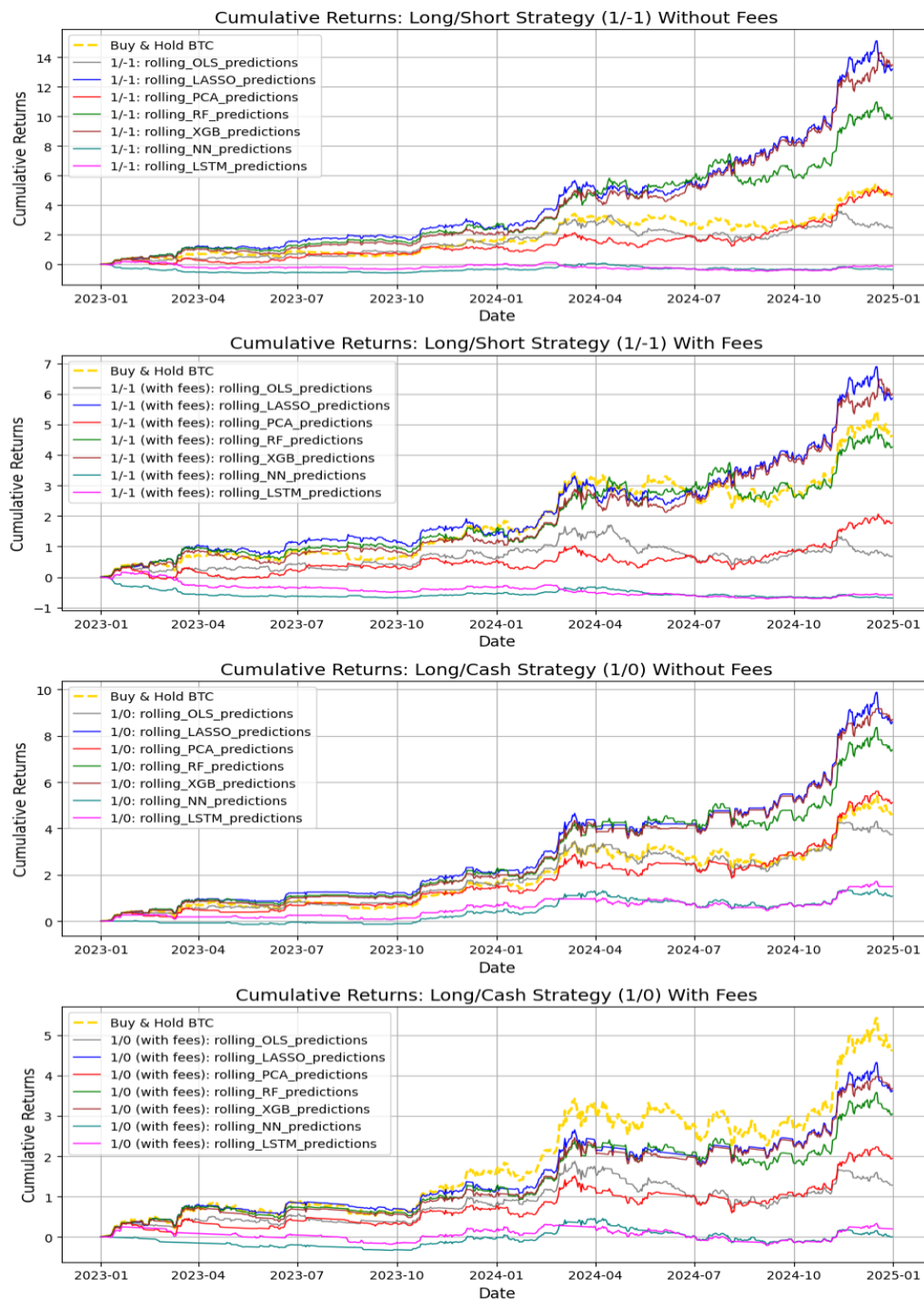


Figure 2 - Cont'd

Panel B: 2022 - 2024

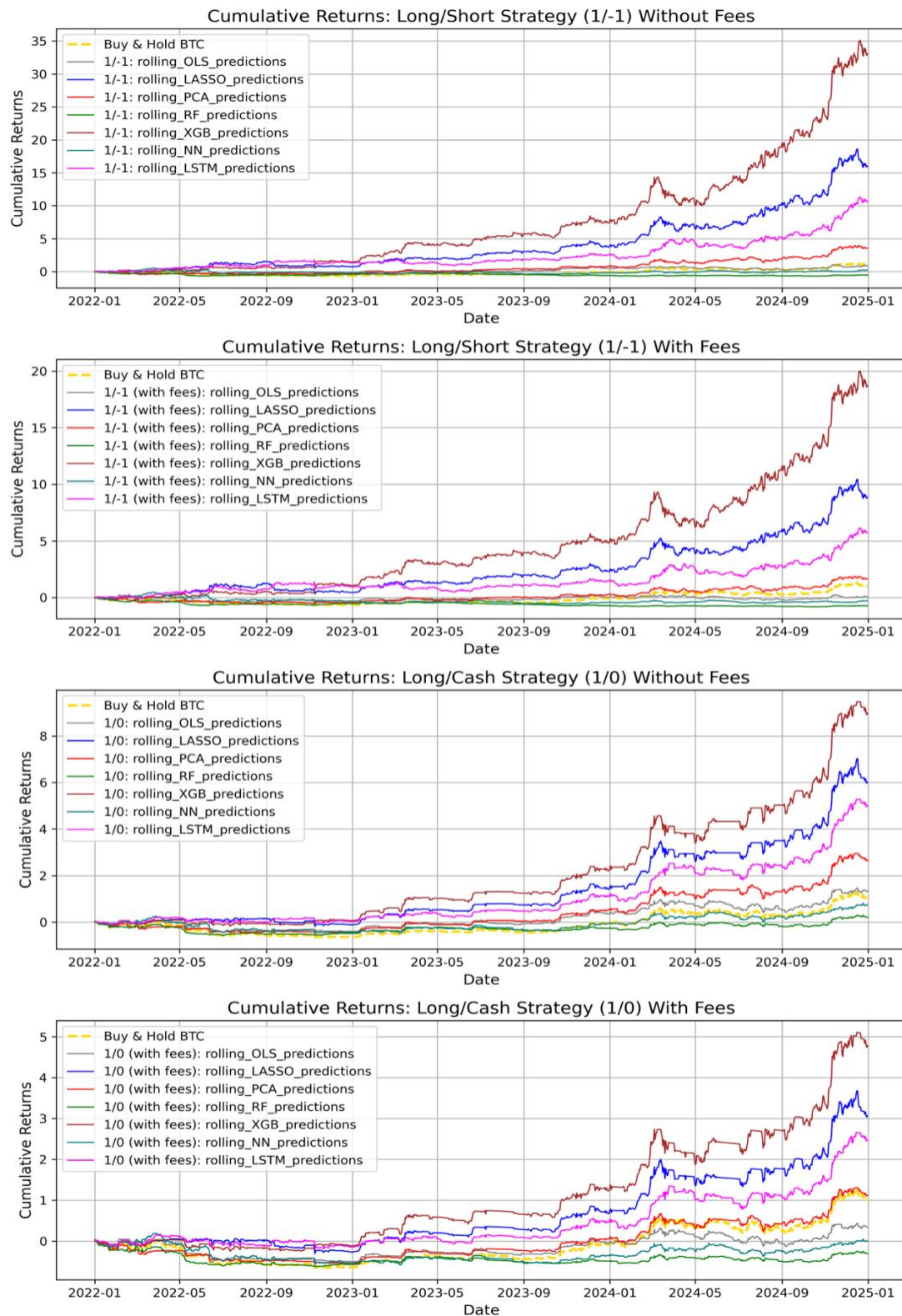


Table 1: Summary statistics

This table reports sample size, mean, standard deviation, and selected quantiles for daily Bitcoin log returns and a representative subset of predictors from each type (market, macro, sentiment, on-chain). All variables are lagged one day for forecasting. The full list of variables and summary statistics appears in Appendix A.

	N	Mean	Std	Min	P25	Median	P75	Max
Market indicators								
Log Market Cap	4383	24.95	2.28	18.76	22.77	25.59	26.99	28.37
Log Volume	4383	10.55	1.97	0.00	9.87	10.85	11.70	14.15
Volume Shock 30D	4383	0.03	0.71	-2.21	-0.34	-0.02	0.32	8.57
Momentum 30D	4383	0.00	0.01	-0.03	0.00	0.00	0.01	0.06
Turnover	4383	0.42	0.07	0.00	0.40	0.43	0.46	0.58
Price MA Signal Short	4383	0.46	0.50	0.00	0.00	0.00	1.00	1.00
Economic indicators								
S&P 500 Return	4383	0.00	0.01	-0.12	0.00	0.00	0.00	0.09
VIX Return	4383	0.00	0.07	-0.28	-0.02	0.00	0.01	1.16
Gold Return	4383	0.00	0.01	-0.09	0.00	0.00	0.00	0.06
US Dollar Index Return	4383	0.00	0.00	-0.02	0.00	0.00	0.00	0.02
FED Funds	4383	1.50	1.82	0.05	0.09	0.54	2.27	5.33
2-year Treasury	4383	1.73	1.56	0.09	0.43	1.21	2.63	5.19
10-year Treasury	4383	2.47	0.97	0.52	1.81	2.37	2.93	4.98
Sentiment Indicators								
BTC Google Trend	4383	19.48	16.88	0.00	7.72	13.90	26.04	124.06
Fear And Greed Index	4383	48.23	16.73	5.00	41.00	50.00	50.00	95.00
On-chain indicators								
Inflow Total	4383	10.76	10.48	0.20	10.07	10.67	11.06	12.82
Outflow Total	4383	10.74	10.46	0.00	10.04	10.66	11.05	12.73
Log Addresses Count Inflow	4383	10.45	1.69	3.69	10.26	11.09	11.39	12.86
Log Addresses Count Outflow	4383	10.09	2.02	0.00	9.54	10.89	11.36	12.98
Transactions Count Total	4383	12.46	11.88	10.31	12.03	12.47	12.68	13.73
Log Miner2Exch Flow Total	4383	6.41	1.92	0.00	5.83	6.59	7.37	10.94
Log Exch2Miner Flow Total	4383	5.68	1.95	0.00	5.16	6.04	6.82	10.61
Log Blockreward USD	4383	15.80	1.56	10.64	14.39	16.31	17.08	18.49
Log Reserve USD	4383	22.43	3.41	6.15	19.73	23.68	25.16	26.44
Log Realized Cap	4383	24.44	2.26	18.19	22.26	25.20	26.68	27.42
Profit Percent	4383	78.25	17.76	14.54	69.91	81.53	92.70	100.00
Loss Percent	4383	21.71	17.78	0.00	7.29	18.45	30.09	85.44

Table 2: Forecast error metrics across models

Out of sample one-step ahead performance from a rolling estimation scheme. We report RMSE (Root Mean Squared Error), MSRE (mean squared relative error), MAE (Mean Absolute Error), and MAPE (Mean Absolute Percentage Error) where applicable. Lower values indicate better forecasts. Panel A reports the results for the 2023-24 period; Panel B reports the results for the 2022-24 period.

<i>Panel A: 2023-24</i>							
Model	RMSE	Model	MSRE	Model	MAE	Model	MAPE
XGB	0.0256	XGB	15.68	XGB	0.0180	XGB	155.59
LASSO	0.0256	PCA	21.96	LASSO	0.0180	RF	170.64
PCA	0.0257	RF	22.50	PCA	0.0184	LASSO	170.96
LSTM	0.0260	LASSO	42.37	LSTM	0.0185	PCA	190.45
RF	0.0264	NN	102.72	NN	0.0188	NN	216.17
NN	0.0264	OLS	168.21	RF	0.0188	LSTM	220.89
OLS	0.0293	LSTM	3330.03	OLS	0.0216	OLS	609.15
<i>Panel B: 2022-24</i>							
Model	RMSE	Model	MSRE	Model	MAE	Model	MAPE
XGB	0.0285	XGB	18.11	XGB	0.0197	XGB	151.45
LASSO	0.0286	LASSO	47.85	LASSO	0.0197	LASSO	196.69
PCA	0.0288	LSTM	61.11	PCA	0.0200	PCA	217.79
NN	0.0293	PCA	85.76	NN	0.0204	LSTM	230.09
LSTM	0.0293	NN	122.44	LSTM	0.0205	NN	239.75
RF	0.0299	RF	199.42	RF	0.0211	RF	274.16
OLS	0.0315	OLS	2362.68	OLS	0.0229	OLS	541.15

Table 3: Out of sample R^2 by model

Incremental explanatory power relative to the historical-mean benchmark. Positive values indicate improvement over the benchmark. Results are based on rolling one-day-ahead forecasts.

<i>Panel A: 2023-24</i>		<i>Panel B: 2022-24</i>	
Model	R^2_{oos}	Model	R^2_{oos}
LASSO	0.0118	XGB	0.0134
XGB	0.0103	LASSO	0.0104
PCA	0.0027	PCA	-0.0040
LSTM	-0.0164	NN	-0.0375
RF	-0.0481	LSTM	-0.0386
NN	-0.0509	RF	-0.0839
OLS	-0.2970	OLS	-0.2000

Table 4: Directional accuracy: Bernoulli test

This table reports prediction stats across all models. % + *Predictions* and % - *Predictions* are the shares of days the model predicts an up or down return. *Correct + Predictions %* (*Correct - Predictions %*) is the hit rate conditional on the day actually being up (down). *Total % Correct* is overall sign accuracy. *P-values* are from a two-sided exact binomial test of the null that accuracy equals 50%.

<i>Panel A: 2023-24</i>							
Model	Actual % + Returns	% + Predictions	% - Predictions	Correct + Pred. (%)	Correct - Pred. (%)	Total % Correct	P-value
<i>XGB</i>	50.62	61.15	38.85	55.70	57.39	56.36	0.0003
<i>LASSO</i>	50.62	63.20	36.80	55.41	57.62	56.22	0.0004
<i>PCA</i>	50.62	59.92	40.08	53.20	53.24	53.21	0.0444
<i>LSTM</i>	50.62	55.27	44.73	53.47	52.91	53.21	0.0444
<i>OLS</i>	50.62	69.08	30.92	51.49	51.33	51.44	0.2297
<i>NN</i>	50.62	74.69	25.31	50.00	47.57	49.38	0.6442
<i>RF</i>	50.62	56.50	43.50	49.39	47.80	48.70	0.7703
<i>Panel B: 2022-24</i>							
Model	Actual % + Returns	% + Predictions	% - Predictions	Correct + Pred. (%)	Correct - Pred. (%)	Total % Correct	P-value
<i>XGB</i>	49.09	57.39	42.61	53.74	57.17	55.20	0.0003
<i>LASSO</i>	49.09	58.21	41.79	53.45	56.99	54.93	0.0006
<i>LSTM</i>	49.09	56.39	43.61	52.75	55.65	54.01	0.0043
<i>PCA</i>	49.09	50.64	49.36	51.89	53.79	52.83	0.0327
<i>OLS</i>	49.09	59.58	40.42	49.77	51.92	50.64	0.3473
<i>NN</i>	49.09	74.91	25.09	49.21	51.27	49.73	0.5837
<i>RF</i>	49.09	53.10	46.90	47.94	49.61	48.72	0.8095

Table 5: Diebold–Mariano test matrix

Pairwise tests of equal predictive accuracy using squared-error loss on out-of-sample one-day-ahead forecasts. Entries report DM statistics (or p-values, as labeled). Positive statistics favor the column model. Asterisks denote significance (***, **, * at 1%, 5%, 10%).

	<i>Panel A: 2023-24</i>						
	<i>OLS</i>	<i>LASSO</i>	<i>PCA</i>	<i>RF</i>	<i>XGB</i>	<i>NN</i>	<i>LSTM</i>
<i>OLS</i>		4.22***	4.11***	3.40***	4.24***	3.28***	3.82***
<i>LASSO</i>	-4.22***		-0.82	-3.36***	0.18	-3.22***	-1.90*
<i>PCA</i>	-4.11***	0.82		-2.79***	0.82	-2.76***	-1.33
<i>RF</i>	-3.40***	3.36***	2.79***		3.82***	-0.13	1.46
<i>XGB</i>	-4.24***	-0.18	-0.82	-3.82***		-4.11***	-1.86*
<i>NN</i>	-3.28***	3.22***	2.76***	0.13	4.11***		1.58
<i>LSTM</i>	-3.82***	1.90*	1.33	-1.46	1.86*	-1.58	

	<i>Panel B: 2022-24</i>						
	<i>OLS</i>	<i>LASSO</i>	<i>PCA</i>	<i>RF</i>	<i>XGB</i>	<i>NN</i>	<i>LSTM</i>
<i>OLS</i>		4.68***	4.37***	2.32**	4.72***	3.50***	3.47***
<i>LASSO</i>	-4.68***		-1.62	-4.19***	0.34	-3.07***	-3.26***
<i>PCA</i>	-4.37***	1.62		-3.72***	1.57	-1.98**	-2.16**
<i>RF</i>	-2.32**	4.19***	3.72***		4.38***	1.80*	1.77*
<i>XGB</i>	-4.72***	-0.34	-1.57	-4.38***		-4.17***	-3.41***
<i>NN</i>	-3.50***	3.07***	1.98**	-1.80*	4.17***		-0.06
<i>LSTM</i>	-3.47***	3.26***	2.16**	-1.77*	3.41***	0.06	

Table 6: Sharpe ratios for BTC-equity allocation strategies

This table reports annualized Sharpe ratios for daily two-asset allocations between Bitcoin (BTC) and the equity market. To align with equity trading days, BTC's returns between equity closes are geometrically compounded into a single daily BTC return for that trading day. Portfolio weights each trading day are proportional to each sleeve's forecast scaled by its recent risk: the BTC sleeve uses the model's forecasted BTC return, and the market sleeve uses its own mean return forecast; specifically, we form unscaled weights $g_t^{BTC} = \hat{r}_{t+1}^{BTC} / (\sigma_t^{BTC})^2$ and $g_t^{Mkt} = \bar{r}_{t+1}^{Mkt} / (\sigma_t^{Mkt})^2$, apply box constraints $[-0.5, 1.5]$ applied to each sleeve and weights renormalized to sum to 1. The Sharpe ratio is computed as: $[Sharpe = \frac{\bar{r}_t - r_{ft}}{\hat{\sigma}(r_t - r_{ft})} \sqrt{252}]$ where r_t is the daily strategy return and r_{ft} the risk-free rate; 252 reflects equity trading days. "Buy and Hold" is buy-and-hold BTC. "Market returns" is buy-and-hold equity.

Panel A: 2023-24		Panel B: 2022-24	
Model	Sharpe Ratios	Model	Sharpe Ratios
<i>LASSO</i>	2.1846	<i>XGB</i>	0.4512
<i>Buy and hold</i>	1.8289	<i>Buy and hold</i>	0.4487
<i>XGB</i>	1.7176	<i>LASSO</i>	0.3268
<i>PCA</i>	1.7015	<i>Market returns</i>	0.3194
<i>NN</i>	1.4841	<i>NN</i>	0.0422
<i>Market returns</i>	1.4140	<i>LSTM</i>	0.0147
<i>RF</i>	1.4036	<i>OLS</i>	-0.0128
<i>LSTM</i>	1.3096	<i>PCA</i>	-0.1932
<i>OLS</i>	1.0561	<i>RF</i>	-0.2052

Appendix A: Variables definition

Variable	Source	Definition
<i>Panel A: Macro Indicators</i>		
Crude Oil WTI Momentum	Author's computation	Indicator variable equal to 1 if the 7-day moving average of the underlying index is greater than the 30-day moving average, 0 otherwise.
Crude Oil WTI Returns	Yahoo Finance	Daily return on Crude Oil WTI index (CL=F).
Crude Oil WTI Volume	Yahoo Finance	Natural logarithm of 1 plus the number of contracts traded during the day.
DGS2	FED (FRED)	Daily yield on 2-year U.S. Treasury securities at constant maturity (FRED DGS2 series).
DGS2 delta	Author's computation	Change in DGS2
DGS10	FED (FRED)	Daily yield on 10-year U.S. Treasury securities at constant maturity (FRED DGS10 series).
DGS10 delta	Author's computation	Change in DGS10
EUR-USD Momentum	Author's computation	Indicator variable equal to 1 if the 7-day moving average of the underlying index is greater than the 30-day moving average, 0 otherwise.
EUR-USD Returns	Yahoo Finance	The rate of change on Euro-Dollar exchange rate.
FEDFUNDS	FED (FRED)	Effective federal funds rate (average overnight lending rate between depository institutions). (FRED FEDFUNDS series).
FEDFUNDS delta	Author's computation	Change in FEDFUNDS
Gold Momentum	Author's computation	Indicator variable equal to 1 if the 7-day moving average of the underlying index is greater than the 30-day moving average, 0 otherwise.
Gold Returns	Yahoo Finance	Daily return on front-month Comex Gold Futures (GC=F).
Gold Volume	Yahoo Finance	Natural logarithm of 1 plus the number of contracts traded during the day.
JPY-USD Momentum	Author's computation	Indicator variable equal to 1 if the 7-day moving average of the underlying index is greater than the 30-day moving average, 0 otherwise.
JPY-USD Returns	Yahoo Finance	The rate of change on Yen-Dollar exchange rate.

MSCI World ETF Momentum	Author's computation	Indicator variable equal to 1 if the 7-day moving average of the underlying index is greater than the 30-day moving average, 0 otherwise.
MSCI World ETF Returns	Yahoo Finance	Daily return on iShares MSCI World ETF (URTH).
MSCI World ETF Volume	Yahoo Finance	Natural logarithm of 1 plus the number of ETF shares traded during the day.
R_EG	HXZ Q-factors	Q-factor: expected growth factor returns per Hou et al., (2021)
R_F	HXZ Q-factors	Q-factor: one-month Treasury bill rates per Hou et al., (2021)
R_IA	HXZ Q-factors	Q-factor: investment factor returns per Hou et al., (2021)
R_ME	HXZ Q-factors	Q-factor: size factor returns per Hou et al., (2021)
R_MKT	HXZ Q-factors	Q-factor: market excess returns per Hou et al., (2021)
R_ROE	HXZ Q-factors	Q-factor: return on equity factor returns per Hou et al., (2021)
S&P 500 Momentum	Author's computation	Indicator variable equal to 1 if the 7-day moving average of the underlying index is greater than the 30-day moving average, 0 otherwise.
S&P 500 Returns	Yahoo Finance	Daily return on the S&P 500 index (^GSPC).
S&P 500 Volume	Yahoo Finance	Natural logarithm of 1 plus the total number of shares traded across all 500 stocks shares during the day.
US Dollar Index Momentum	Author's computation	Indicator variable equal to 1 if the 7-day moving average of the underlying index is greater than the 30-day moving average, 0 otherwise.
US Dollar Index Return	Yahoo Finance	The rate of change of the US Dollar Index (DX-Y.NYB).
VIX Momentum	Author's computation	Indicator variable equal to 1 if the 7-day moving average of the underlying index is greater than the 30-day moving average, 0 otherwise.
VIX Returns	Yahoo Finance	The rate of change of the VIX index.
WALCL	FED (FRED)	Natural logarithm of 1 plus the Total assets held by the Federal Reserve (in \$M) (FRED WALCL series).
WALCL delta	Author's computation	Change in WALCL

Panel B: Market Indicators

Close	CryptoQuant	Price of Bitcoin at the end of the trading day (23:59 UTC).
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High	CryptoQuant	Highest recorded Bitcoin price during the trading day.
Illiq	Author's computation	Log-transformed <i>Amihud (2002)</i> illiquidity ratio, approximating price impact per unit of volume.
Illiq 7DMA	Author's computation	7-day moving average of <i>Illiq</i> .
Intraday HI	Author's computation	Daily price range, computed as the difference between the high and low price.
Kurtosis	Author's computation	Rolling 90-day kurtosis of daily BTC returns.
Lag Returns	Author's computation	Daily BTC returns lagged by one day (P_{t-1} / P_{t-2}) - 1
Lag2 Returns	Author's computation	Daily BTC returns lagged by two days (P_{t-2} / P_{t-3}) - 1
Lag3 Returns	Author's computation	Daily BTC returns lagged by three days (P_{t-3} / P_{t-4}) - 1
Lag4 Returns	Author's computation	Daily BTC returns lagged by four days (P_{t-4} / P_{t-5}) - 1
Lag5 Returns	Author's computation	Daily BTC returns lagged by five days (P_{t-5} / P_{t-6}) - 1
Monday	CryptoQuant	Indicator variable equal to 1 if the observation occurs on a Monday, 0 otherwise.
Low	CryptoQuant	Lowest recorded Bitcoin price during the trading day.
Market Cap	CryptoQuant	Natural logarithm of 1 plus BTC's daily market capitalization.
Max	CryptoQuant	Maximum BTC returns observed over the prior 30 trading days.
Min	CryptoQuant	Minimum BTC returns observed over the prior 30 trading days.
Momentum 7D	Author's computation	7-day rolling average of daily BTC returns; captures short-term return.
Momentum 30D	Author's computation	30-day rolling average of returns; reflects longer-term return momentum.
Momentum 6M	Author's computation	180-day rolling average of returns, lagged by 30 days; represents delayed six-month momentum.
Open	CryptoQuant	Price of Bitcoin at the beginning of the trading day (00:00 UTC).

Price MA 30	Author's computation	30-day price moving average.
Price MA Signal Med	Author's computation	Dummy variable equal to 1 if the 7-day moving average of BTC price is greater than or equal to the 28-day moving average, 0 otherwise.
Price MA Signal Short	Author's computation	Dummy variable equal to 1 if the 3-day moving average of BTC price is greater than or equal to the 15-day moving average.
Realized Vol 30D	CryptoQuant	30-day rolling standard deviation of returns as in <i>Yang and Zhang (2000)</i> .
Realized Vol 180D	CryptoQuant	180-day rolling standard deviation of returns as in <i>Yang and Zhang (2000)</i> .
Reversal 2D	CryptoQuant	2-day rolling average of returns; measures very short-term mean reversion or reversal.
Reversal 1M	CryptoQuant	30-day rolling average of returns; a 1-month mean reversion measure.
Reversal 1Y	CryptoQuant	365-day rolling average of returns, lagged by 30 days; captures delayed one-year reversal effects as in <i>Cong et al. (2022)</i> .
Skewness	Author's computation	Rolling 90-day skewness of daily BTC returns.
Std Turnover 30D	Author's computation	30-day rolling standard deviations of turnover.
Std Turnover 180D	Author's computation	180-day rolling standard deviations of turnover, capturing volume volatility.
Std Vol 30D	Author's computation	30-day rolling standard deviations of log volume, capturing volume volatility.
Std Vol 180D	Author's computation	180-day rolling standard deviations of log volume.
Saturday	Author's computation	Indicator variable equal to 1 if the observation occurs on a Saturday, 0 otherwise.
Sunday	Author's computation	Indicator variable equal to 1 if the observation occurs on a Sunday, 0 otherwise.
Thursday	Author's computation	Indicator variable equal to 1 if the observation occurs on a Thursday, 0 otherwise.
Tuesday	Author's computation	Indicator variable equal to 1 if the observation occurs on a Tuesday, 0 otherwise.
Turnover	Author's computation	Ratio of log volume to log market cap.
Volume	CryptoQuant	Natural logarithm of 1 plus daily BTC trading volume.

Volume Shock 30D	Author's computation	Deviation of log volume from its 30-day rolling average, as in <i>Llorente et al. (2002)</i> .
Volume Shock 60D	Author's computation	Deviation of log volume from its 60-day rolling average as in <i>Llorente et al. (2002)</i> .
Wednesday	Author's computation	Indicator variable equal to 1 if the observation occurs on a Wednesday, 0 otherwise.
<i>Panel C: Sentiment Indicators</i>		
BTC Google trend	CryptoQuant	Google Search Bitcoin Trend index.
BTC Google trend delta	Author's computation	Change in <i>BTC Google trend</i> .
Fear and Greed Index	CoinMarketcap	A CoinMarketCap proprietary index ranging from 0 to 100 that measures the prevailing sentiment in the cryptocurrency market.
Fear and Greed Index delta	Author's computation	Change in <i>Fear and Greed index</i> .
<i>Panel D: On-Chain Indicators</i>		
Addresses Count Inflow	CryptoQuant	Natural logarithm of 1 plus the number of addresses involved in inflow transactions to crypto-exchanges.
Addresses Count Outflow	CryptoQuant	Natural logarithm of 1 plus the number of addresses involved in outflow transactions out of crypto-exchanges.
Addresses Count Receiver	CryptoQuant	Natural logarithm of 1 plus the number of addresses that were active as a receiver on the Bitcoin blockchain.
Addresses Count Sender	CryptoQuant	Natural logarithm of 1 plus the number of addresses that were active as a sender on the Bitcoin blockchain.
Average Cap	CryptoQuant	A forever moving average computed as the Natural logarithm of 1 plus the cumulated sum of daily market cap divided by the age of the market.
Binary CDD	CryptoQuant	Indicator variable equal to 1 if supply adjusted CDD is above its average value, and 0 otherwise.
Block Bytes	CryptoQuant	Natural logarithm of 1 plus the mean size (in bytes) of all blocks generated.
Block Bytes delta	Author's computation	Change in <i>Block Bytes</i> .
Block Count	CryptoQuant	The number of blocks generated that day.
Block Interval	CryptoQuant	The average time between blocks generated on the day of observation (in seconds).

Block Interval delta	Author's computation	Change in <i>Block Interval</i> .
Blockreward Usd	CryptoQuant	Natural logarithm of 1 plus the sum of block rewards (in \$).
CDD	CryptoQuant	Coin Days Destroyed (CDD) is the sum of the product of coin amounts and the number of days since they were last moved. High CDD values indicate movement of long-dormant coins, often associated with long-term holder activity.
Days Since Halving	Author's computation	Number of days since last halving event.
Delta Cap	CryptoQuant	Natural logarithm of 1 plus (Realized Cap – Average Cap). Useful tool to spot market bottoms.
Difficulty	CryptoQuant	Natural logarithm of 1 plus the mean difficulty of mining a new block.
Exch2Miner Flow Total	CryptoQuant	Natural logarithm of 1 plus the total number of tokens transferred from a <i>crypto-exchange</i> to a <i>Mining Pool</i> .
Exch2Miner Transactions Count Flow	CryptoQuant	Natural logarithm of 1 plus the total number of transactions transferred from a <i>crypto-exchange</i> to a <i>Mining Pool</i> .
Exchange Supply Ratio	CryptoQuant	Exchange reserves divided by total supply.
Exchange Whale Ratio	CryptoQuant	The ratio of the top 10 inflows to the total inflows on all crypto-exchanges. High values indicate whales are using the exchanges in large amount.
Fees Reward Percent	CryptoQuant	The percentage of fees relative to the total block reward on average on the day of observation.
Fees block Mean USD	CryptoQuant	The average fee per block (in \$).
Fees Transaction Mean USD	CryptoQuant	The average fee per transaction in each block (in \$).
Fees Transaction Mean USD delta	Author's computation	Change in <i>Fees Transaction Mean USD</i>
Fund Flow Ratio	CryptoQuant	The ratio of the amount of bitcoins that exchanges occupy among the bitcoins sent underlying the Bitcoin network.
Hashrate	CryptoQuant	Natural logarithm of 1 plus the mean speed at which miners in the network are solving hash problems (in hashes per second).
Hashrate delta	CryptoQuant	Change in <i>Hashrate</i> .
Inflow CDD	CryptoQuant	Measures the age-weighted volume of coins flowing into exchanges. It reflects the cumulative dormant time of coins being deposited.
Inflow Total	CryptoQuant	Natural logarithm of 1 plus the inflow of BTC into <i>exchange wallets</i> .

Is Halving Day	CryptoQuant	Indicator variable equal to 1 if observation occurs on the day of a halving event, and 0 otherwise.
Loss Amount	CryptoQuant	The dollar amount of UTxOs being in loss or not by comparing the price between created and destroyed.
Loss Percent	CryptoQuant	The percentage of UTxOs being in loss or not by comparing the price between created and destroyed.
MCDA	CryptoQuant	Mean Coin Dollar Age (MCDA) is the ratio of CDD divided by the total moved coin.s
MF Addresses Count Inflow	CryptoQuant	Natural logarithm of 1 plus the number of addresses involved in inflow transactions to <i>Mining Pool wallets</i> .
MF Addresses Count Outflow	CryptoQuant	Natural logarithm of 1 plus the number of addresses involved in outflow transactions from <i>Mining Pool wallets</i> .
MF Inflow Mean	CryptoQuant	Natural logarithm of 1 plus the average inflow of BTC into <i>Mining Pool wallets</i> during the day of observation.
MF Inflow Mean delta	Author's computation	Change in <i>MF Inflow Mean</i> .
MF Outflow Mean	CryptoQuant	Natural logarithm of 1 plus the average outflow of BTC out of <i>Mining Pool wallets</i> during the day of observation.
MF Outflow Mean delta	Author's computation	Change in <i>MF Outflow Mean</i> .
MF Reserve Usd	CryptoQuant	Natural logarithm of 1 plus the on-chain balance of Bitcoin in Mining Pools (in \$).
MF Transactions Count Inflow	CryptoQuant	Natural logarithm of 1 plus the number of transactions flowing into <i>Mining Pool wallets</i> .
MF Transactions Count Inflow delta	Author's computation	Change in MF Transactions Count Inflow.
MF Transactions Count Outflow	CryptoQuant	Natural logarithm of 1 plus the number of transactions flowing out of <i>Mining Pool wallets</i> .
MF Transactions Count Outflow delta	Author's computation	Change in MF Transactions Count Outflow.
Miner2Exch Flow Total	CryptoQuant	Natural logarithm of 1 plus the total number of tokens transferred from a <i>Mining Pool</i> to a <i>crypto-exchange</i> .
Miner2Exch Transactions Count Flow	CryptoQuant	Natural logarithm of 1 plus the total number of transactions transferred from a <i>Mining Pool</i> to a <i>crypto-exchange</i> .
MPI	CryptoQuant	Miner's Position Index (MPI): the ratio of total miner outflow (\$) to its one-year moving average of total miner outflow (\$).

NUL	CryptoQuant	<i>Net Unrealized Loss</i> (NUL), which is calculated as sum of products of negative UTxO's value and price difference between created and destroyed, divided by market cap.
NUP	CryptoQuant	<i>Net Unrealized Profit</i> (NUP), which is calculated as sum of products of positive UTxO's value and price difference between created and destroyed, divided by market cap.
NVT	CryptoQuant	<i>Network Value to Transaction</i> (NVT) ratio is the dollar market cap divided by the total token transferred.
NVT Golden Cross	CryptoQuant	The difference between the 10-day moving average NVT and the 30-day moving average NVT divided by NVT 300-day volatility.
Outflow Total	CryptoQuant	Natural logarithm of 1 plus the outflow of BTC out of exchange wallets.
Profit Amount	CryptoQuant	The dollar amount of UTxOs being in profit or not by comparing the price between created and destroyed.
Profit Percent	CryptoQuant	The percentage of UTxOs being in profit or not by comparing the price between created and destroyed.
Puell Multiple	CryptoQuant	The mining revenue (in \$) divided by 365-day moving average mining revenue (in \$).
Realized Cap	CryptoQuant	Natural logarithm of 1 plus the sum of each UTXO * last movement price.
Reserve USD	CryptoQuant	Natural logarithm of 1 plus the full historical on-chain balance of Bitcoin on crypto-exchanges (in \$).
Reserve USD delta	Author's computation	Change in Reserve USD.
SCA	CryptoQuant	Sum Coin Age is the sum value of products of bitcoin unspent transaction output times age.
Stock To Flow	CryptoQuant	The ratio of currently circulating coins divided by newly supplied coins.
Supply New	CryptoQuant	Natural logarithm of 1 plus the amount of newly issued tokens on the day.
Supply Total	CryptoQuant	Natural logarithm of 1 plus the total amount of bitcoins in existence (sum of all bitcoins issued by the coinbase reward).
Thermo Cap	CryptoQuant	Natural logarithm of 1 plus the weighted sum of mined coins by the creation price.
Tokens Transferred Mean	CryptoQuant	Natural logarithm of 1 plus the mean of number of transferred tokens per transaction.
Tokens Transferred Total	CryptoQuant	Natural logarithm of 1 plus the total number of transferred tokens.
Transactions Count Flow	CryptoQuant	Natural logarithm of 1 plus the total number of transactions flowing in/out of crypto-exchanges.

Transactions Count Inflow	CryptoQuant	Natural logarithm of 1 plus number of transactions flowing to crypto-exchanges.
Transactions Count Mean	CryptoQuant	Natural logarithm of 1 plus the mean number of transactions on the Bitcoin blockchain on the day of observation.
Transactions Count Outflow	CryptoQuant	Natural logarithm of 1 plus number of transactions flowing out of crypto-exchanges.
Transactions Count Total	CryptoQuant	Natural logarithm of 1 plus total number of transactions on the Bitcoin blockchain on day of observation.
UTXO Count	CryptoQuant	Natural logarithm of 1 plus the total number of unspent transaction outputs existing on the day of the observation.
UTXO Count delta	Author's computation	Change in UTXO Count.
Velocity Supply Total	CryptoQuant	The trailing 1 year estimated transaction volume divided by current supply.

Appendix B: Hyperparameter search settings

Model	CV design	Search procedure	Settings
LASSO <i>sklearn</i>	5-fold, block time ordered	Grid search	- Alpha $\in \{1e-5, 1e-4, 1e-3, 0.01, 0.1, 1, 10\}$ - Max_iter = 5000
PCA	Recomputed each window	N/A	- Standardize features; smallest k with $\geq 90\%$ cumulative variance - N_estimators [10, 50, 100, 200]
Random Forest <i>sklearn</i>	2-fold time ordered	RandomizedSearchCV, n_iter = 20	- Max_features = 'sqrt' - Max_depth [10, 20, 30, 40, 50] - Min_samples_split [2, 5, 10] - Min_samples_leaf [1, 2, 4] - N_estimators [50, 100, 200] - Max_depth [3, 5, 10] - Learning_rate [0.01, 0.05, 0.1]
XGBoost <i>XGBRegressor</i>	2-fold time ordered	RandomizedSearchCV, n_iter = 25	- Subsample [0.7, 0.9, 1.0] - Colsample_bytree [0.7, 0.9, 1.0] - Reg_alpha [0, 0.1, 1] - Reg_lambda [0, 0.1, 1] - Tree_method = 'hist' - Hidden layers: 3 - Units per layer $\in [32 \dots 256]$ step 32 - Activations {relu, tanh, leaky_relu} - BatchNorm + Dropout $\in [0.1 \dots 0.5]$
NN <i>KerasTuner</i>	Chronological 80/20 split inside window	RandomSearch (max_trials = 15, executions_per_trial = 2)	- Per-layer L1/L2 $\in [1e-4 \dots 1e-2]$ (log) - Optimizers {adam, sgd, rmsprop} - LR $\in [1e-4 \dots 1e-2]$ (log) - Batch_size $\in \{16, 32, 48, 64\}$ - Epochs $\in \{5, 10, 15, 20\}$ - Output linear - 3 LSTM layers (final feeds Dense(1))
LSTM <i>KerasTuner</i>	Chronological 80/20 split inside window	RandomSearch (max_trials = 15, executions_per_trial = 2)	- Units: L1 $\in \{50, 100, 150, 200\}$, L2 $\in \{50, 100, 150\}$, L3 $\in \{50, 100\}$ - Dropout $\in [0.1 \dots 0.5]$ - Optimizers {adam, rmsprop, sgd} - LR $\in [1e-4 \dots 1e-2]$ (log) - Training epochs = 8, Batch_size = 32

Unless otherwise noted, all models are trained to minimize Mean Squared Error (MSE) on daily returns and are tuned using cross-validation performed strictly within each rolling training window. The CV design column specifies how the window is split: blocked, time-ordered means folds are contiguous in time and each validation fold occurs after its training fold; chronological 80/20 split means the last 20% of the window is used as validation with no shuffling. All preprocessing (standardization and, when used, PCA) is fit on the training portion only of each split and then applied to the corresponding validation portion to prevent look-ahead leakage. For PCA we retain the smallest number of components that explains $\geq 90\%$ of variance, recomputed in each window.

For tree models, RandomizedSearchCV samples hyperparameter combinations using the stated CV design; for neural networks, KerasTuner RandomSearch explores the architecture/optimizer space with an in-window chronological validation split and early stopping (patience = 3, restore best weights). After selecting hyperparameters on the inner CV, each model is refit on the full training window and then used to issue the one-day-ahead forecast. Seeds are fixed for reproducibility (sklearn/XGBoost random_state=42; Keras inherits the process seed). All splits preserve chronology (no shuffling).